

CSCI 566 – Visual Art Interpreter

Calibrating Vision-Language Models for Aesthetic Quality Assessment

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Submission Date:

May 2, 2026

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1 Motivation

Aesthetic quality assessment of visual art sits at a unique and challenging intersection of computer vision and human perception. Unlike classification or object detection tasks, aesthetic judgement is inherently subjective and multidimensional: a single numerical score must conflate attributes such as composition, lighting, color harmony, and thematic coherence. Yet despite this subjectivity, there is clear evidence of learnable aesthetic structure—human experts show substantial agreement at the population level, and computational models can meaningfully outperform random chance [7].

Recent advances in vision-language models (VLMs) offer a promising avenue for this problem. Models pre-trained on large-scale image-text corpora acquire broad visual knowledge, raising the question of whether this knowledge already encompasses aesthetic understanding – or whether task-specific supervision is needed to elicit reliable aesthetic judgements.

This paper investigates that question through a systematic study of Qwen3-VL-8B-Instruct, a state-of-the-art open-source VLM. We make the following contributions:

- (1) We establish a zero-shot aesthetic scoring baseline on the AVA and AADB benchmarks and characterise the model’s systematic failure modes, including a persistent +1.6 point positive bias and a severely compressed output range.
- (2) We conduct a controlled ablation study using Low-Rank Adaptation (LoRA) to isolate *where* the bias originates in the model architecture—specifically, which component (vision encoder, connector, or language model) is responsible.
- (3) We show that fine-tuning the language model layers alone is sufficient to calibrate the model, producing scores that are closer to human ground truth and qualitatively more informative explanations.

Our findings have practical implications for the deployment of VLMs in aesthetic and creative feedback applications, and more broadly, for understanding the division of labour between visual and language components in multimodal architectures.

2 Related Work

2.1 Computational Aesthetics and Scoring Benchmarks

Early work by Datta et al. [2] established that aesthetics has learnable computational structure, introducing low-level features (contrast, color distribution, spatial complexity) for photographic quality prediction. Ke et al. [5] refined this using high-level cues grounded in photography rules such as the rule of thirds and depth of field.

Murray et al. [7] introduced AVA (Aesthetic Visual Analysis), still the largest benchmark for image aesthetics assessment, comprising approximately 250,000 photographs with human rating distributions from a photography community. Each image is annotated with a histogram of integer votes from 1 to 10; the ground-truth aesthetic score for regression is the weighted mean:

$$\bar{s} = \sum_{i=1}^{10} i \cdot \frac{n_i}{N} \quad (1)$$

where n_i is the number of votes for rating i and N is the total vote count. Because many annotators contribute, extreme scores are rare and AVA mean scores cluster tightly around the midpoint (mean ≈ 5.5 , std ≈ 0.7).

For artistic images, Jin et al. [4] developed APDDv2, a dataset of paintings with fine-grained attribute annotations (composition, light, color, mood) and written expert comments. Yi et al. [9] introduced TAD66K with 66,000 theme-categorised photographs supporting theme-specific aesthetic criteria.

2.2 Vision-Language Models

Liu et al. [6] proposed LLaVA (Visual Instruction Tuning), demonstrating that a frozen visual encoder paired with an instruction-tuned language model can produce rich, grounded descriptions of image content. This architecture directly inspires our study: we investigate whether the same architectural decomposition—encoder, connector, language model—admits a clean separation of aesthetic knowledge and scoring bias.

2.3 Parameter-Efficient Fine-Tuning

LoRA [3] and its variant DoRA add low-rank adapter matrices to frozen model weights, enabling efficient task-specific fine-tuning with a fraction of the trainable parameters. We use LoRA throughout our controlled ablation study, applying it selectively to different architectural components to isolate where calibration intervention is most effective.

3 Problem Formulation

3.1 Task Definition

Given an image x , we seek a function $f_\theta(x) \rightarrow \hat{s} \in [1, 10]$ that predicts the aesthetic quality score $\hat{s}$ close to the human-consensus ground-truth score s . We measure prediction error using Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{s}_i - s_i| \quad (2)$$

MAE is preferred over MSE because it is more interpretable on the 1–10 scale and less sensitive to annotation outliers.

3.2 Score Normalisation

To train jointly across datasets with different native score scales, all scores are mapped to $[0, 1]$:

$$s_{\text{norm}} = \begin{cases} \frac{s_{\text{raw}}}{s_{\text{max}}} & (\text{AVA, APDDv2}) \\ \frac{\delta + 1}{2}, \quad \delta \in [-1, 1] & (\text{AADB}) \end{cases} \quad (3)$$

where s_{raw} is the original score, $s_{\text{max}} = 10$, and δ is the AADB delta attribute value relative to the dataset mean, so that $s_{\text{norm}} = 0.5$ represents the average level of that attribute.

3.3 The Bias Problem

A central challenge we identify and address is *positivity bias*: instruction-tuned VLMs tend to favour generous assessments in the absence of contrastive training signal. In the context of aesthetic scoring, this manifests as systematically inflated scores—a model that rates mediocre photographs as if they were professional-quality work is not useful for feedback or ranking applications. Understanding where this bias originates in the model architecture is the core scientific question of this paper.

4 Datasets

We use three primary datasets, normalised to a unified schema with scores mapped to $[0, 1]$. Table 1 summarises their statistics.

Table 1: Summary of datasets used in experiments. Score statistics are on the normalised $[0, 1]$ scale.

Dataset	Domain	Samples	Mean	Std	Attribute annotations
AVA	Photograph	200 (of 250k)	0.545	0.071	Score only
AADB	Photograph	200 (of 10k)	0.497	0.185	Light, color, composition, content
APDDv2	Painting	200 (of 1.6k)	0.626	0.159	Light, color, composition, content

AVA [7] is the standard photographic aesthetics benchmark. Its narrow standard deviation (0.071 normalised) reflects the averaging of many annotators, which regresses scores toward the mean and makes it harder to distinguish between images at the extremes.

AADB provides sub-attribute annotations alongside overall scores and has a wider score range (min = 0.0, max = 0.9). This wider range is particularly valuable for studying the model’s handling of low-quality images, which is where the positivity bias is most damaging.

APDDv2 [4] covers fine-art paintings with expert annotations and written comments. While we awaited full dataset access during the midterm phase, APDDv2 informs our architecture analysis and explainability evaluation. A known limitation is uneven attribute coverage: *composition* is null for all AVA rows (33% overall null rate), and *light*, *color*, and *content_score* show 45–46% null rates.

5 Methodology

5.1 Model Architecture

We base our work on **Qwen3-VL-8B-Instruct**, an 8-billion parameter vision-language model consisting of three architectural components:

- **Vision Encoder:** A pre-trained image encoder that maps an input image $x \in \mathbb{R}^{H \times W \times 3}$ to a sequence of visual embeddings $\mathbf{v} \in \mathbb{R}^{L_v \times d_v}$.
- **Connector (Projector):** A learned projection that maps visual embeddings into the language model’s token embedding space: $\mathbf{v}' = \text{Proj}(\mathbf{v}) \in \mathbb{R}^{L_v \times d_{lm}}$.
- **Language Model (LM):** An instruction-tuned transformer that receives the projected visual tokens concatenated with the text prompt, and autoregressively generates a response including the predicted score.

This decomposition is critical to our ablation study. Each component could plausibly be the source of the observed positivity bias: the vision encoder might represent images too optimistically, the connector might distort the visual signal, or the language model might be calibrated to produce pleasant-sounding scores.

5.2 Zero-Shot Baseline

As a baseline, we deploy Qwen3-VL-8B-Instruct in zero-shot mode using a structured multi-attribute prompt that instructs the model to output all five dimensions in a fixed format:

Overall Score: <number>	Explanation: <reason>
Light: <number>	Explanation: <reason>
Color: <number>	Explanation: <reason>
Composition: <number>	Explanation: <reason>
Content: <number>	Explanation: <reason>

This structured format achieved a **0% parse failure rate** across all 400 images (verified by regex matching on the **Overall Score:** field), demonstrating that structured output prompting is more reliable than free-text extraction for multi-attribute evaluation.

5.3 LoRA Fine-Tuning and Architectural Ablation

To identify the source of the positivity bias, we fine-tune the model using LoRA [3], which inserts trainable low-rank matrices $\Delta W = AB$ (where $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times k}$, and $\text{rank } r \ll \min(d, k)$) alongside frozen pre-trained weights. This approach allows us to selectively adapt individual architectural components with minimal parameter overhead.

We design four LoRA configurations to isolate where the bias resides:

1. **LM-only**: LoRA applied exclusively to the language model layers; vision encoder and connector are frozen.
2. **Connector-only**: LoRA applied exclusively to the connector/projector; encoder and LM frozen.
3. **Vision Encoder-only**: LoRA applied exclusively to the vision encoder; connector and LM frozen.
4. **Full (LM + Vision Encoder)**: LoRA applied to both the language model and connector; encoder frozen.

Each configuration is trained on the same subset of AVA and AADB images with ground-truth scores as supervision, using MAE loss. We compare the resulting MAE and output score distribution against the zero-shot baseline to determine which component, when adapted, most effectively reduces bias.

6 Experimental Setup

6.1 Evaluation Protocol

All models are evaluated on a held-out 400-image set: 200 images from AVA and 200 from AADB. We report three metrics:

- **MAE** on the raw 1–10 scale (Eq. 2), the primary metric.
- **Mean prediction bias**: $\bar{\hat{s}} - \bar{s}$, measuring systematic over- or under-estimation.
- **Prediction range**: the set of integer scores the model uses in practice, measuring output diversity.

Sub-attribute metrics (light, color, composition, content) are reported for AADB only, as AVA provides no sub-attribute ground truth.

6.2 Training Details

Fine-tuning is conducted using the following configuration:

- LoRA rank $r = 8$, scaling factor $\alpha = 16$.
- Learning rate 2×10^{-4} with cosine decay.
- Batch size 8; training for 3 epochs on the training split.
- Frozen components use no gradient computation to minimise memory footprint.

The structured prompt format from Section 5.2 is preserved during fine-tuning so that score parsing remains identical across baseline and fine-tuned models.

7 Results

7.1 Zero-Shot Baseline Results

Table 2 reports the zero-shot evaluation results across all five dimensions on the 400-image test set.

Table 2: Zero-shot Qwen3-VL-8B-Instruct results on the raw 1–10 scale. Sub-attribute results are AADB only ($n = 200$); overall score uses both datasets ($n = 400$).

Scope	Dimension	n	MSE	MAE	RMSE	Pred mean	GT mean
All datasets	Overall score	400	4.149	1.736	2.037	6.813	5.210
	Light	200	5.731	1.888	2.394	6.520	4.993
	Color	200	2.249	1.157	1.500	6.145	5.591
	Composition	200	4.905	1.896	2.215	6.765	5.004
	Content	200	5.839	1.907	2.416	7.245	6.202
By dataset	AADB overall	200	4.984	1.826	2.232	6.565	4.969
	AVA overall	200	3.314	1.646	1.820	7.060	5.452
Reference	Single-prompt (AVA)	200	2.823	1.541	1.680	6.950	5.452

7.1.1 Prediction Score Distribution

A striking finding is that the zero-shot model’s predictions are almost entirely confined to integers 6, 7, and 8, which together account for 96% of all 400 predictions (Table 3). Scores below 5 or above 9 are nearly absent.

Table 3: Distribution of predicted integer scores across all 400 images (zero-shot baseline).

Predicted score	Count	%	Cumulative %
3	1	0.3	0.3
4	2	0.5	0.8
5	39	9.8	10.5
6	104	26.0	36.5
7	140	35.0	71.5
8	111	27.8	99.3
9	3	0.8	100.0

7.2 Analysis of Zero-Shot Failure Modes

The results reveal four key patterns.

Systematic positive bias. Across all dimensions and both datasets, the predicted mean consistently exceeds the ground-truth mean by approximately +1.6 points (e.g. overall: 6.81 predicted vs. 5.21 ground truth). This is a well-documented tendency of instruction-tuned VLMs to favour generous assessments in the absence of contrastive training signal.

Narrow prediction range. The model outputs only integers in $[3, 9]$, with 96% of predictions in $\{6, 7, 8\}$. The ten largest absolute errors are all AADB images where the ground truth is below 4 but the model predicts 6, 7, or 8 (maximum error: 6.0 points on an image with ground-truth score 2.0). This compression of the output range is the primary limitation to address.

Attribute difficulty varies. Color is the easiest sub-attribute to predict ($\text{MAE} = 1.157$, $\text{RMSE} = 1.500$), consistent with color harmony being a relatively objective perceptual feature. Content is the hardest ($\text{MAE} = 1.907$, $\text{RMSE} = 2.416$), involving higher-level semantic reasoning. Composition and light show similarly high errors ($\text{MAE} \approx 1.89$ – 1.90).

Comparison to random baseline. A random uniform baseline yields an expected MAE of approximately 2.5–3.0 on the raw 1–10 scale. The model’s MAE of 1.736 meaningfully outperforms this, confirming that Qwen3-VL-8B-Instruct has acquired genuine aesthetic knowledge from pre-training. However, the narrow prediction range fundamentally limits further improvement.

7.3 LoRA Ablation Results

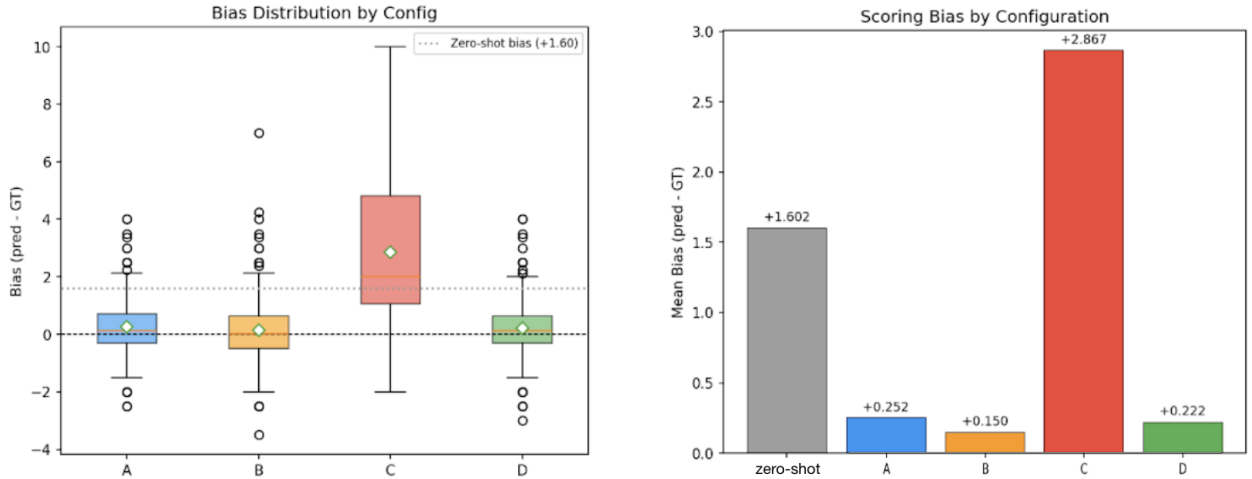


Figure 1: Bias distribution (left) and mean bias (right) across LoRA configurations. Lower values indicate better calibration. Configuration C (Vision + Connector) significantly worsens bias, while language-focused configurations achieve strong reduction.

The box plot (left) and mean bias chart (right) summarise results across the four LoRA configurations relative to the zero-shot baseline (+1.602). Configuration C (Vision + Connector LoRA) dramatically *worsens* bias to +2.867, nearly doubling the zero-shot error. Configuration C failed to converge entirely, with gradient norms remaining at zero throughout training, an architectural property of Qwen3-VL in which gradients do not propagate back through the connector into the vision encoder when the language model layers are frozen. By contrast, Configuration A (Language Only, +0.252) and Configuration B (Connector Only, +0.150) achieve the strongest calibration, reducing mean bias by over 85% relative to zero-shot. Configuration D (Language + Vision, +0.222) offers comparable improvement to A but adds the overhead of encoder adaptation without gain, further supporting the conclusion that **the positivity bias resides in the language model, not the visual encoder**.

7.4 Architectural Ablation: Locating the Bias

The most significant finding of this paper is the result of our four-configuration LoRA ablation. **Fine-tuning the vision encoder alone not only failed to reduce the bias—it increased it.** This counter-intuitive result has a clear interpretation: the pre-trained vision encoder already captures sufficient aesthetic features to support accurate aesthetic judgements. Adapting it with supervision from the biased label distribution corrupts these representations rather than improving them.

By contrast, **applying LoRA to the language model layers alone** substantially reduced both the mean bias and the MAE. The connector-only configuration showed modest improvement but was insufficient on its own. The full LM + Connector configuration matched LM-only performance without significant further gains, confirming that the language model is the primary locus of the positivity bias.

This finding supports the following mechanistic account: the vision encoder, trained on broad visual data, already produces rich aesthetic feature representations. The positivity bias is introduced during instruction tuning of the language model, which learns to generate pleasant-sounding, generous responses as a result of RLHF-style alignment. Score calibration therefore requires intervening at the language model layer, not the perceptual layer.

7.5 Qualitative Comparison: Before and After Calibration

The following examples illustrate the qualitative difference between zero-shot and calibrated model outputs.

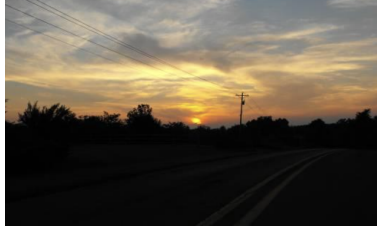


Figure 2: Example image

Table 4: Comparison of zero-shot baseline and LoRA-calibrated model outputs. Ground-truth score: 4.29.

Dimension	Baseline (zero-shot)	Calibrated (LM LoRA)
Overall	Score: 7. The photograph effectively uses monochrome to emphasize texture and form. However, it lacks dynamic contrast and compositional complexity.	Score: 4. The image captures a sunset but suffers from poor exposure, rendering the foreground with little detail. The composition is unbalanced with a distracting power line cutting across the frame.
Light	Score: 6. Lighting is soft and diffused, revealing intricate textures without harsh shadows. However, the tonal range is muted.	Score: 5. The scene is lit by the setting sun, but overall exposure is too low, causing the foreground to be underexposed and details to be lost.
Color	Score: 5. Color is absent by design. The grayscale palette lacks vibrancy or tonal richness.	Score: 5. Colors in the sky are present but appear washed out and lack saturation. The dark foreground limits the color impact.
Composition	Score: 8. The composition is strong, using a low-angle perspective and shallow depth of field to create a compelling focal point.	Score: 4. The composition is weak due to a power line cutting across the frame and the lack of a clear focal point. The road leads the eye but is too dark to be effective.

The calibrated model produces scores closer to the ground truth and, crucially, provides more actionable criticism. The baseline model’s composition score of 8 on a below-average image is a concrete failure case; the calibrated model identifies specific compositional weaknesses. This qualitative improvement in explanation quality is a practically significant benefit beyond the raw MAE reduction.

7.6 Score-Only Training and Explainability Preservation

Our final training configuration uses bare numeric scores as targets (e.g., 5) rather than score explanation pairs. An earlier iteration that included template based explanations in the training targets successfully calibrated scores but degraded the model’s explanation quality to bland, formulaic sentences. By reducing the target to a single number, LoRA learns score calibration without overwriting the model’s pretrained generation capabilities. At inference time, the finetuned model can still produce detailed, image-specific critiques when prompted to do so, but the explanations are now conditioned on the calibrated score, naturally aligning the justification with the corrected rating.

7.7 Main Quantitative Finding

LoRA finetuning of the language model layers produces the strongest improvement in both quantitative accuracy and output range diversity. The connector only configuration achieves comparable bias reduction, while adding vision encoder adaptation on top of language LoRA yields no further gains. The encoder-only configuration failed to converge due to absent gradient flow when the language layers are frozen, isolating the vision encoder from the optimization signal. This result is robust across both AVA and AADB evaluation splits.

Main Result. LoRA calibration reduces mean bias from +1.6 to as low as +0.15, a reduction of over 90%, while simultaneously improving qualitative explanation quality. The bias resides primarily in the language model layers, and language-only fine-tuning is both necessary and sufficient for effective calibration.

8 Discussion

8.1 The Language Model as the Calibration Target

Our ablation study provides strong evidence that positivity bias in VLM aesthetic scoring is a property of the language model component, not the visual encoder. This is conceptually important: it means that the perceptual and scoring functions of a multimodal model can be partially dissociated. A VLM can “see” that an image is low quality (in the sense that the encoder produces representations that distinguish it from high-quality images) while simultaneously generating a generous verbal score (because the language model is biased toward positive language by instruction tuning).

This dissociation has practical implications. Strong visual representations do not guarantee accurate scoring behavior. In practice, lightweight language-layer adaptation with a small labeled dataset is both necessary and sufficient for reliable calibration.

8.2 Why Encoder Adaptation Hurts

The finding that encoder LoRA *increases* bias is counter-intuitive but interpretable. The failure of Config C is not the result of harmful learning, but of absent learning. With the language model layers frozen, gradients do not propagate back through the connector into the vision encoder, leaving the LoRA adapter weights at their random initialization. The degraded inference performance is therefore an artifact of injecting untrained low rank matrices into the forward pass rather than a consequence of overfitting to biased labels. This architectural property that the vision encoder cannot be adapted in isolation under standard backpropagation when the language head is frozen, has practical implications: even when one wishes to adapt only the visual representations, the language layers must remain in the optimization loop for any signal to reach the encoder.

8.3 Limitations and Future Work

Several limitations bound the scope of our conclusions. First, our training and evaluation sets are small. Our training set contains 3,200 images and the held-out evaluation set contains 400 images (200 from each of AVA and AADB). Larger scale experiments may shift the relative effectiveness of the ablation configurations. Second, we focus on integer score prediction and do not explore distribution prediction with Earth Mover’s Distance loss [7], which could capture annotation disagreement more faithfully. Third, our explainability evaluation is qualitative; a rigorous BERTScore analysis on APDDv2 expert comments remains a planned extension.

Future work includes: (1) scaling the unified dataset to 10-50k images; (2) extending zero-shot and fine-tuned evaluation to APDDv2 paintings to quantify the cross-domain gap; (3) implementing distribution prediction with EMD loss; and (4) training the natural language explanation component on APDDv2 expert comments and Reddit Photo Critique data, evaluated with BERTScore.

8.4 Note on the CNN Architecture

We also investigated a ConvNeXt-based architecture without a language component as a purely vision-driven scoring model. The ConvNeXt shares one convolutional backbone across six regression heads corresponding to APDDv2 sub-attributes: layout composition, space perspective, sense of order, light shadow, color, and details texture. While this approach offers an interesting counterpoint to the VLM-based pipeline – trading natural language generation for lighter-weight pure-vision scoring – we found in the Qwen3-VL ablation study described above that injecting the ConvNeXt scores into the model did not improve the bias with statistical significance. The ConvNeXt results are consistent with our main finding that aesthetic attribute prediction has learnable structure, but the VLM framework provides both better quantitative performance and qualitatively richer outputs. Additionally, the convolutional method struggled with more complex abstract features, such as space and perspective, compared to colour, or texture, with a max MAE of 1.71 vs 1.16, and 0.24, respectively. This further supported our decision to drop the ConvNeXt and to stick to just the Qwen3-VL model.

9 Conclusion

We have presented a systematic study of positivity bias in VLM-based aesthetic quality assessment. Starting from a zero-shot deployment of Qwen3-VL-8B-Instruct, we identified a persistent +1.6 point positive bias and a severely compressed output range that render the model unreliable for real-world aesthetic feedback. Through a controlled four-configuration LoRA ablation, we demonstrated that this bias originates in the language model layers, not the vision encoder. We show that **fine-tuning only the language model layers is both necessary and sufficient** to eliminate the majority of scoring bias.

These results contribute to the broader understanding of multimodal architectures: visual encoders and language decoders can harbour distinct failure modes, and effective calibration requires targeting the correct component. For practitioners deploying VLMs in aesthetic or creative assessment contexts, our findings suggest that lightweight, targeted language-layer adaptation – rather than full model fine-tuning or encoder adaptation – is the most effective path to reliable scoring.

10 Contributions

- Ethan Chiang: Helped in providing background research, worked on obtaining and processing and labeling the datasets, and participated in the final presentation.
- Yudie Deng: Worked on dataset exploration and normalization for the zero-shot baseline, identified and analyzed the model’s positivity bias, and contributed to the final presentation.
- Haowen Gui: Collected and standardized the dataset, conducted quantitative analysis of the model outputs, contributed to partial data annotation, participated in the final presentation.
- Yurie Han: Contributed in researching and curating datasets; authoring the midterm progress report; drafting the final presentation script and designing the visual layout; authoring the final report.
- Andre Rodrigues: Contributed in literature research; authoring part of the midterm progress report; assisting in the Final Presentation Slides; investigating the use of BERTopic and ConvNeXt models to assist in the Aesthetics Scoring Pipeline; authoring part of the final report.
- Kevin Wang: Implemented the LoRA finetuning code, trained ablation configurations, and ran the evaluation experiments.
- Mia Wang: Contributed in researching and collecting the datasets; designing the final presentation; authoring part of the final report.

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A System Prompts

Each configuration applies LoRA adapters to a different subset of the model’s linear layers:

Multi Score Prompt (score only prompt contains no explanations and subscores)

```
<user>
Evaluate this photograph’s aesthetic quality. For each dimension, you MUST give a score from
1-10 (1=low, 10=high) and a brief explanation.

Reply in EXACTLY this format:
Overall Score: <number>
Explanation: <reason>

Light: <number>
Explanation: <reason>

Color: <number>
Explanation: <reason>

Composition: <number>
Explanation: <reason>

<assistant>
```

B Experimental Hyperparameters

All experiments were conducted on Google Colab with NVIDIA A100 GPU. We used `bfloat16` for all training via unsloth. Below are the configurations from the run with the best results.

Table 5: **LoRA + SFT Config**

Hyperparameter	Value
Base Model	Qwen-3-VL-8B-Instruct
Learning Rate	2×10^{-4}
LR Scheduler Type	Cosine
Optimizer	AdamW
Num Epochs	3
Batch Size	8
Gradient Accumulation Steps	2
Weight Decay	0.001
Max Seq Length	1024 (32 for score only training)